MILITARY SPECIFICATION

ELECTRONIC COMPONENT ASSEMBLY: 11738818 (CHARGE CONTROL)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electronic component assembly known as Component Assembly: 11738818 (Charge Control) (see 6. 1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment; Acquisition, Maintanance, and Disposition of
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication, and Radar for Aerospace and Control Systems, Procedure for

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MIL-S-46844	Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Tape

STANDARDS

Military

MIL-STD-135	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1O9	Quality Assurance Terms and Definitions
MIL-STD-81O	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F11738816	Electronic Component Assembly; 11738818 (Charge Control)

PACKAGIG DATA SHEET

11738818	Packaging of Electronic Component Assembly; 11738818 (Charge Control)

(Copies of specifications. standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The electronic component assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing F11738818 and drawings pertaining thereto.

3.2 General specification. The assembly shall meet the following requirements, where applicable, of MIL-F-13926:

a. Order of precedence

b. Dimensions and tolerances

c. Part ldentification and marking

d. Electrical and electronic assemblies

e. Workmanship

3.2.1 Soldering. Soldering shall be in accordance with MIL-S-45743 or MIL-S-46844, except that bare copper wire exposure shall be permitted at clipped ends of component leads.

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (°F) and 90°F.

3.3.1 Loads, power, signal, and interconnections. The assembly shall perform as specified herein when the loads, power, signal, and interconnections of table I are applied as specified.

3.3.1.1 Low/high battery voltage charge current. With the power source (item 2.1) of table I applied to P1-24 and adjusted to +24.0 ±2.0 Volts direct current (Vdc), and with the load (item 1.1) of table I connected, the current flowing through the specified load shall be 200 +80 milliamperes (mA). With the load (item 1.1) of table I disconnected and with the load (item 1.2) of table I connected, the current flowing through the specified load shall be 200 ±8O mA.

3.3.1.2 Charger voltage regulation. With the power source (item 2.1) of table I applied to F1-24 and adjusted to +24.0 ±2.0 Vdc, and with the load (item 1.3) of table I connected as specified. the voltage of P1-4 shall be +43.O ±4.0 Vdc. With the above conditions existing and with P1-15 connected to P1-16, the voltage of P1-4 shall be +36.0 ±3.0 Vdc.

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TABLE I
LOADS, POWER , SIGNAL, AND INTERCONNECTIONS

Item	Conditions	Characteristics	Connections
1.	Loads		Connected between the following pins:
1.1	Resistor	51 ohms (Ω) ±5 percent	P1-4(+) and P1-7(-) when specified herein
1.2	Resistor	130Ω ±5 percent	P1-4(+) and P1-7(-) when specified herein
1.3	Resistor	300Ω ±5 percent	P1-4(+) and P1-7(-) when specified herein
1.4	Capacitor	100µf ±20 percent @75 Vdc working voltage	P1-4(+) and P1-7(-)
1.5	Resistor	10 kΩ ±5 percent	P1-8(+) and P1-7(-)
2.	Power sources		
2.1	+17 to +31 Vdc	Adjustable Ripple: ≤0.1 Vrms	Applied between P1-24(+) and P1-7(-) except as specified herein.
2.2	+5 Vdc	Tolerance: ±0.2 Vdc Ripple: ≤25mVp-p	Between P1-12(+) and P1-7(-)
2.3	-6Vdc	Tolerance: ±0.2 Vdc Ripple: ≤25MVp-p	Between P1-11(-) and P1-7(+)
3.	Signal Source		
3.1	Voltage source	Waveform as shown in figure 1. Source impedance 100Ω ± 5 percent	Applied to P1-5, and P1-6 as shown in figure 1
4.	Interconnections		P1-15 and P1-16 when specified herein. P1-10 and P1-7

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3.3.1.3 Input voltage /charge current regulation. With the Power source (item 2.1) of table I applied to P1-24 and adjusted to +18.0 ± 0.5 Vdc and then +30.0 ± 0.5 Vdc, and the load (item 1.2) of table I connected as specified, the current flowing through the specified load shall be 200 +80 mA for each input condition.

3.3.1.4 Charger sense voltage. With the power source (item 2.1) of table I applied to P1-24 and adjusted to +24.0 ±2.0 Vdc, and with the load (item 1.2) of table I connected as specified, the voltage at P1-8 shall be equal to the voltage at PI-4 within +1.5 Vdc.

3.3.1.5 Charge enable signal. With the power source (item 2.1) of table I applied and adjusted to +24.0 ±2.0 Vdc, with the load (item 1.2) of table I connected as specified, and with P1-10 disconnedted, from, P1-7, the current flowing through the specified load shall be 0 +10 mA.

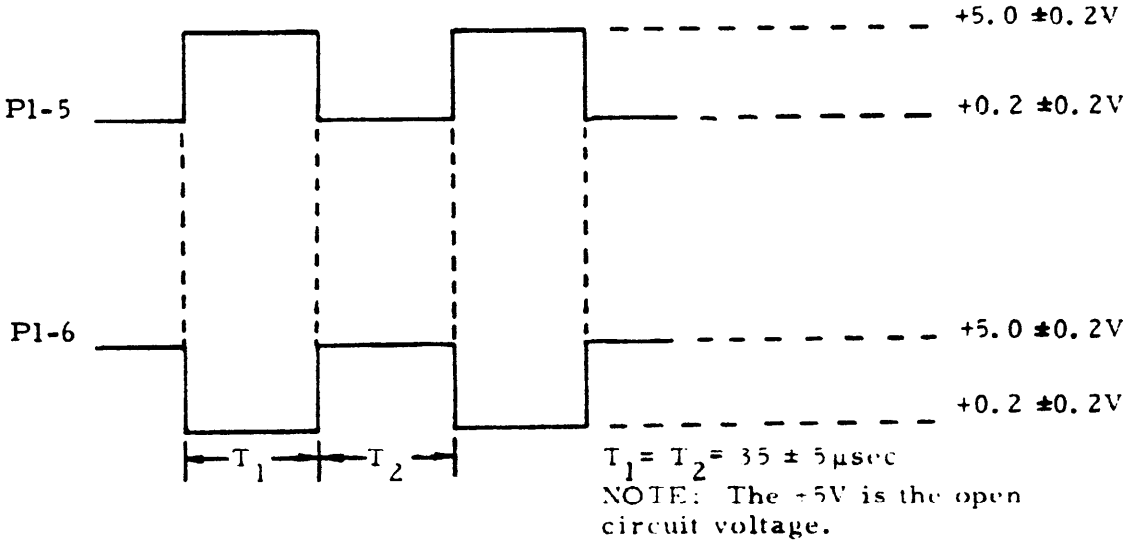


Figure 1. Converter Drive Signal Waveforms

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3.3.2 Environmental

3.3.2.1 Low temperature. The assembly shall meet the requirements of 3.3.1 at ambient temperature after exposure to and thermal stabilization at -80°F and +l60°F.

3.3.2.2 Operating temperatures. The assembly shall meet the requirements of 3.3.1 v'bile exposed to and thermally stabilized at -40°F and +125°F, subsequent to which the requirements of 3.3.1 shall be met at ambient temperature.

3.3.2.3 Vibration. The assembly shall show no physical failure and shall meet the reauirements or 3.3.1 after being exposed to the following vibratory conditions:

a. Amplitude:	1/64 inch, constant (1/32 inch double plitude)
b. Vibratory motion:	Simple harmonic
c. Sweep time:	5 to 55 Hertz in a 5-minute period
d. Vibration cyclng -	Three linear or logarithmic sweeps applied to each of three mutually perpendicular axes.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of the specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article [initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three assemblies. The sample shall be manufactured in the same manner, using the same materials, equipment processed, and procedures as used in regular Production. All parts and materials including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure conformance to all the requirements of this specification. A record of this inspection, including certificates of conformance for materials, shall be submitted with the sample.

4. 2.2 Contractor testing. When the Government is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of this specification. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor’s inspection.

4.3 Inspection provisions

4.3.1 Submission of product. Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with inspection or sampling procedures herein and submission of product provisions of MIL-STD-105.

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4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the respective specifications and Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP'S, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Functional tests. The requirements and tests in table II shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature.

TABLE II
FUNCTIONAL TESTS

Classification	Characteristic	Requirement	Test Paragraph
101.	Low/high battery voltage charge current	3.3.1.1	4.6.2
102.	Charger voltage regulation	3.3.1.2	4.6.3
103.	Input voltage/charger current regulation	3.3.1.3	4.6.4
104.	Charger sense voltage	3.3.1.4	4.6.5
105.	$\overline{\text{Charge enable}}$ signal	3.3.1.5	4.6.6

4.4 Special sampling. The selection of sample shall be in accordance with section 7 of MIL-STD-105.

4.4.1 General. One assembly shall be selected at random by. a Government representative as a control sample from each 50 produced. The control sample shall meet the requirements and tests in table III.

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TABLE III
SPECIAL SAMPLING - GENERAL

Classi-fication	Control Test	Requirement	Test Paragraph
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specification	3.2	MII -F-13926 - visual

4.4.2 Environmental. Three assemblies shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in table IV.

TABLE IV
SPECIAL SAMPLING - ENVIRONMENTAL

Classi-fication	Control Test	Requirement	Test Paragraph
303.	Storage temperature	3.3.2.1	4.6.7.1
304.	Operating temperature	3.3.2.2	4.6.7.1
305.	Vibration	3.3.2.3	4.6.7.2

4.4.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with the acceptance and rejection criteria specified in MIL-STD-105.

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607. The inspection equipment listed in table V shall be utilized in performing the tests specified herein.

4. 5.1 Government furnished inspection equipment. Where the contract provides for Government furnished inspection equipment, supply and maintenance of inspection equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

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4. 5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAPS forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the test methods and procedures specified in 4.6 of this specification and with the component inspection procedures specified in examination and test facilities provisions of MIL-F-13926. Since tolerance of inspection equipment is normally considered to be within 10 percent of the product tolerance for which intended, this inherent error in the inspection equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of inspection equipment shall be so selected and controlled as to insure that the inspection equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when ono-conforming. Construction shall be such as to facilitate routine calibration of inspection equipment,

4.6 Test methods and procedures

4.6.1 Test conditions. Unless otherwise specified, the test conditions shall be in accordance with the test facilities provisions of MIL-F-13926.

4.6.1.1 Initial conditions. Connect the loads, power source and signal source to the assembly as specified in table I. Unless otherwise specified, set the voltage of the power source specified in table I (item 2.1) to +24.0 ± 2.0 Vdc.

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TABLE V

INSPECTION EQUIPMENT

Item	Nomenclature	Description
1.	DC Power Supply	+17 to +31 Vdc, adjustable Ripple: 0.1 volt rms maximum
2.	DC Power Supply	+5.0 ±0.1 volts dc Ripple: ≤25 mVp-p
3.	DC Power Supply	-6.0 ±0.1 volts dc Ripple: ≤25 mVp-p
4.	DC Voltmeter	0 to +50 volts; accuracy of ±1 percent
5.	DC Milliammeter	0 to 1.0 ampere; accuracy: ±3 percent
6.	Oscilloscope	Frequency response: DC to 50 kHz Input impedance: 1 megohm or greater Sensitivity: 5 mV/cm Sweep time: 0.2 µsec/cm

4.6.2 Low/high battery voltage charge current. Connect the load (items 1.1) as specified in table I. Apply the power source (item 2.1) to PI-24 and adjust tc +24.0 +2.0 Vdc. Measure the load current to determine compliance with 3.3.1.1. Disconnect the load (item 1.1) of table I and connect load (item 1. 2) of table 1. Measure the load current to determine compliance with 3.3.1.1.

4.6.3 Charger voltage regulation. Connect the load (item 1.3) as specified in table I. Apply the power source (item 2.1) to P1-24 and adjust to +24.0 ± 2.0 Vdc. Measure the voltage at P1-4 to determine compliance with 3.3.1.2. Maintaining the above input conditions, connect P1-15 to P1-16. Measure the voltage at P1-4 to determine compliance with 3.3.1.2.

4.6.4 Input voltage/charger current regulation. Connect the load (item 1.2) as specified in table I. Apply the power source (item. 2.1) to PI-24 and adjust to +18.0 ± 0.5 Vdc. Measure the load current to determine compliance with 3.3.1.3. Adjust power source to +30.0 ± 0.5 Vdc. Measure load current to determine compliance with 3.3.1.3.

4.6.5 Charger sense voltage. Connect the load (item 1.2) as specified in table I. Apply the power source (item 2.1) to P1-24 and adjust to +24.0 ± 2.0 Vdc. Measure sure the voltage at P1-8 and P1-4 to determine compliance with 3.3.1.4.

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4.6.6 Charge enable signal. Connect the load (item 1.2) as specified in table I. Disconect P1-10 from P1-7. Apply the power source (item 2.1) to P1-24 and adjust to +24.0 ±2.0 Vdc. Measure the load current to determine compliance with 3.3.1.5.

4.6.7 Environmental tests.

4.6.7 Temperature. The test apparatus shall be in accordance with the applicable test facilities provisions of MIL-F-13926 except that the assembly may be mounted on the inside surface of the test chamber. Perform this test in accordance with Procedure I of MIL-F-13926, with conditions C(-40°F) and E(+125°F), and with the rate of temperature change during cycling not greater than 8°F per minute. At each of the specified operating temperatures, perform the tests of 4.6.2 through 4.6.6 to determine compliance with 3.3.2.1 and 3.3.2.2.

4.6.7.2 Vibration. Perform this test utilizing the apparatus specified in Method 514.1 of MIL-STD-810 and in accordance with Part I of Procedure I therein, except that the vibratory conditions of 3.3.2.3 shall apply. Following vibration exposure, examine the assembly for structural failures, loose hardware, wire breakage, and abrasion. Perform the tests of 4.6.2 through 4.6.6 to determine compliance with 3.3.2.3.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. packaging packing and marking shall be in accordance with Packaging Data Sheet 11733818. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The electronic component assembly is a functional module of the Batterv power Supply PP-6607( ) /VVG-1 l intended for use in the Fire Control Rangefinder AN/VVG-1(). The function of the electronic component assembly is to provide a constant current to recharge the battery power supply nickel-cadmium battery.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Applicable packaging data sheet number. (see 5.1

c. Selection of applicable levels of preservation packaging and packing

d. Applicable stock number

e. Requirement for First Article Submission.

Custodian:

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Preparing Activity:

Army - MU

Project no. 1240-A30

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-E-50739(MU) Electronic Component Assembly: 11738818 (Charge Control)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED.

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